A still image.
A little controlled motion.

Interactive single-image animation using flow-field warping

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From a proposed method to an end-to-end prototype

Scope change, stated explicitly

Original proposal: satellite image upscaling.
Interim and current implementation: user-guided single-image animation.

Current technical rationale: three inspectable image-processing algorithms, direct visual control, and a feasible CPU demonstration.

Now implemented and reviewable

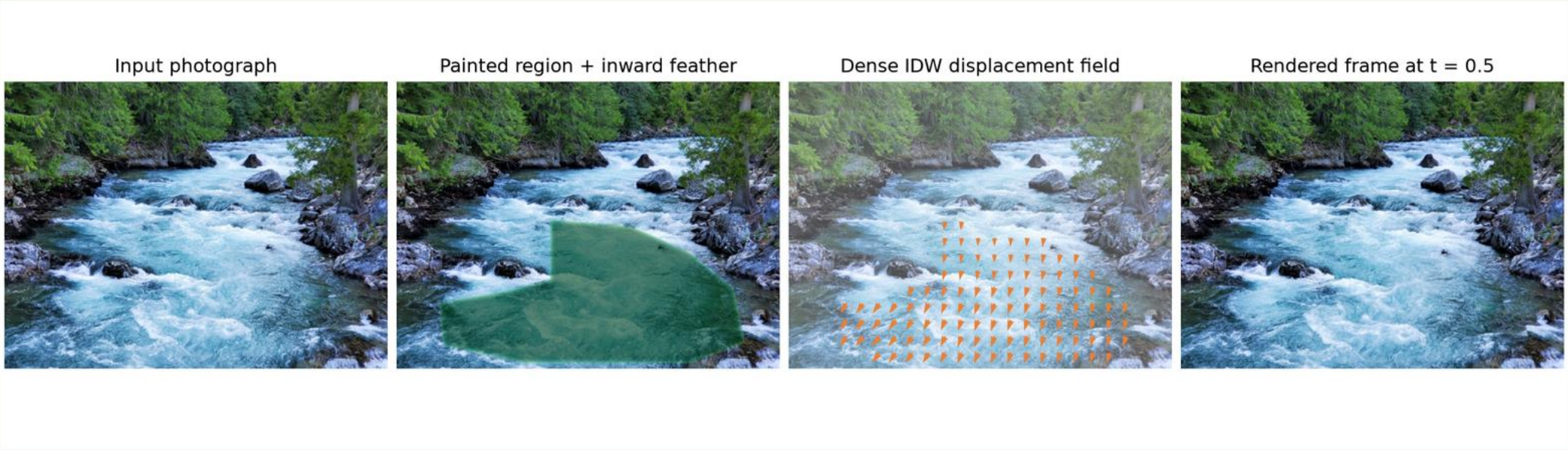
01 Image upload, mask painting and arrows
02 NumPy flow interpolation and warping
03 Loop preview and GIF export
04 Saved projects, examples and tests
05 Reproducible measurements and figures

The whole pipeline is visible

01 Sparse arrows → dense flow

02 Inverse warp + bilinear sample

03 Cross-fade + mask composite



Pillow handles file I/O and GIF encoding. NumPy implements the field, sampling, feathering and frame synthesis.

From a few arrows to moving pixels

1 / Regularized inverse-distance weighting

$$w_i(x) = 1 / (\|x - p_i\|^2 + \varepsilon^2)$$
$$D(x) = \Sigma\, w_i(x)\, a_i / \Sigma\, w_i(x)$$

Nearby arrows have more influence.
ε = 12 pixels prevents singularities.
Zero-motion anchors are soft constraints.
The mask gives exact static protection.

2 / Backward mapping + bilinear sampling

$$A_t(x) = \text{sample}(I, x - tD(x))$$

Four neighboring source pixels are blended by fractional x/y distances. Every destination pixel receives a value.

Clamped borders avoid invalid reads, but can stretch edge textures.

Background: Shepard (1968); Wolberg, Digital Image Warping (1990), inverse mapping and reconstruction.

Make the loop continuous, and keep the background still

$$B_t(x) = \text{sample}(I, x - (t-1)D(x))$$
$$s = \text{clip}((t-1+r)/r, 0, 1)$$
$$\alpha = 3s^2 - 2s^3$$
$$C_t = (1-\alpha)A_t + \alpha B_t$$

$$F_t = I + M(C_t - I)$$
$$M = \min(M_0, \text{GaussianBlur}(M_0))$$

$C_0 = C_1 = I$
Continuous-time endpoint equality

Render t = 0 … (N−1)/N; do not duplicate the endpoint.

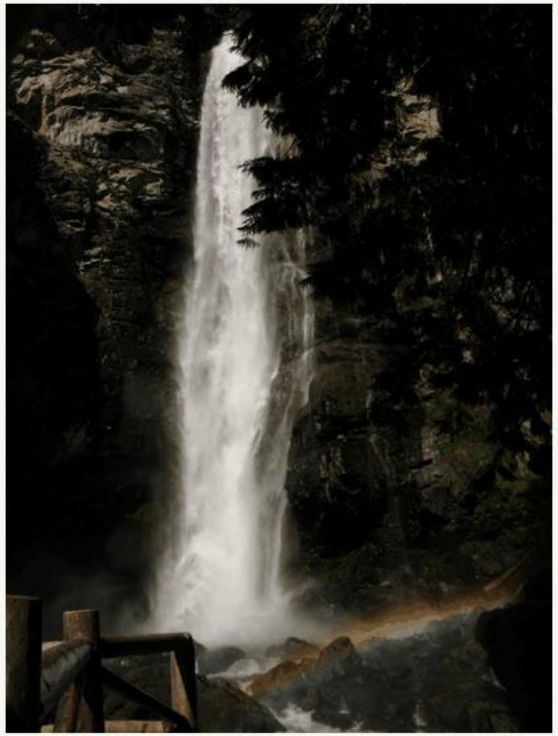
Inward feathering keeps every unpainted RGB pixel unchanged.

Cross-fading can still ghost.

Rendered frames from four reproducible cases



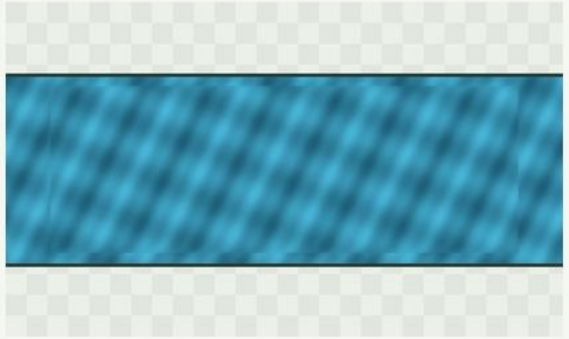
River rapids



Waterfall



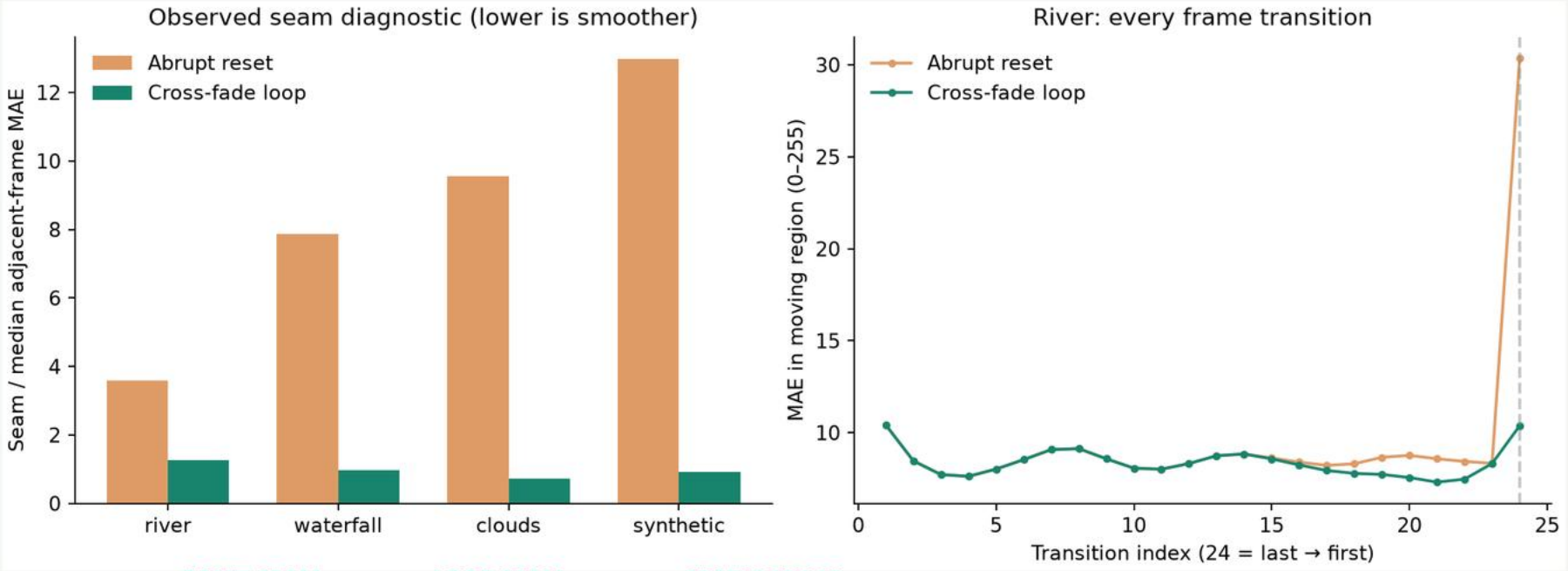
Cloud drift



Procedural test

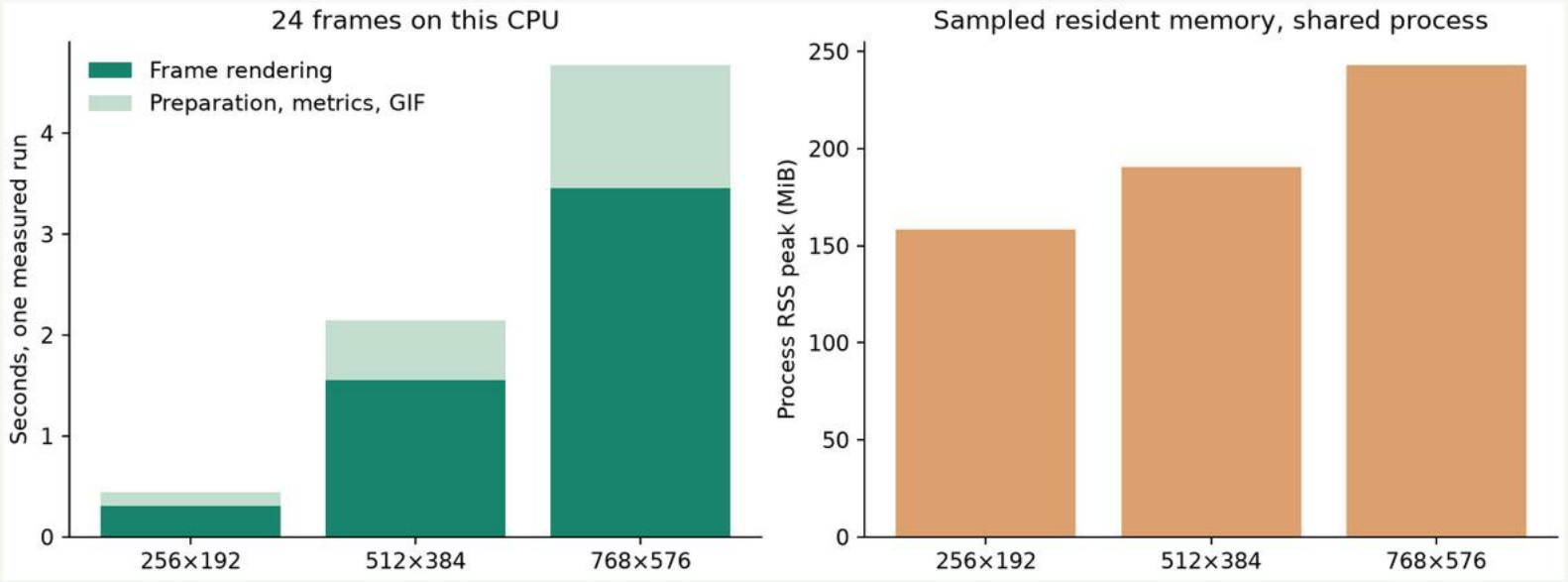
External public-domain NPS photographs; project-created masks and arrows. Original procedural data is also included.

Ablation: the loop boundary is measurably less abrupt



CASE	RESET SEAM	LOOP SEAM	STATIC ERROR
River	3.59×	1.26×	0
Waterfall	7.88×	0.96×	0
Clouds	9.55×	0.72×	0
Synthetic	12.97×	0.91×	0

42 automated tests pass; CPU cost is measured



Correctness checks

Known IDW weights
Bilinear interpolation
Translation direction
Static-pixel invariance
Loop endpoint equality
GIF timing and palette
Upload / render validation

Single-run measurements on the recorded machine. Memory is sampled process RSS, including libraries and retained frames.

What works, what remains limited

Demonstrated

End-to-end user control
Deterministic NumPy algorithms
Repeatable examples and metrics
Exact unpainted-pixel preservation

Current limitations

Ghosting during cross-fades
Stretching for large displacements
No occlusion reconstruction
No physical motion estimation

Next practical work

Add team-captured photographs
Refine masks near rocks / edges
Compare smaller motion settings
Confirm roles and scope rationale

The deliverable is a controllable classical prototype; no new animation theory or human study is claimed.

Live demonstration: image → annotations → loop

01 Open the River rapids example
02 Render, then compare with Original
03 Edit a short arrow or erase a boundary
04 Toggle loop blending for the baseline
05 Download GIF and save the project

Offline fallback

results/demo_river.mp4
results/gifs/river_full.gif
results/gifs/river_no_blend.gif

github.com/Capta7nBlack/robt310-stillflow

References and source credits

[1] Le et al. (2023). Animating Still Natural Images Using Warping. ACM TOMM 19(1), article 4. DOI: 10.1145/3511894.

[2] Shepard (1968). A two-dimensional interpolation function for irregularly-spaced data. ACM National Conference, pp. 517–524. DOI: 10.1145/800186.810616.

[3] Wolberg (1990). Digital Image Warping. IEEE Computer Society Press. §3.1.2 inverse mapping; §5.4.2 linear reconstruction.

[4] Heckbert (1989). Fundamentals of Texture Mapping and Image Warping. UC Berkeley technical report UCB/CSD-89-516.

[5] Photographs: NPS Staff / North Cascades National Park Service Complex; NPS / Cathy Bell. Individual public-domain asset pages are recorded in examples/sources.json.